

Lecture No (2)

Internet Architecture and Protocols

Internet Architecture



Fortunately, nobody owns the Internet, there is no centralized control, and nobody can turn it off. Its evolution depends on rough consensus about technical proposals, and on running code. Engineering feed-back from real implementations is more important than any architectural principles.



Internet

The Internet is a worldwide, publicly accessible network of interconnected computer networks that transmit data by packet switching using the standard Internet Protocol (IP).

It is a "network of networks" A better way for us to start visualizing the Internet is as an opaque cloud to which hosts connect.

The power of the Internet is from respecting the end-to-end principle.



What is the Internet architecture?

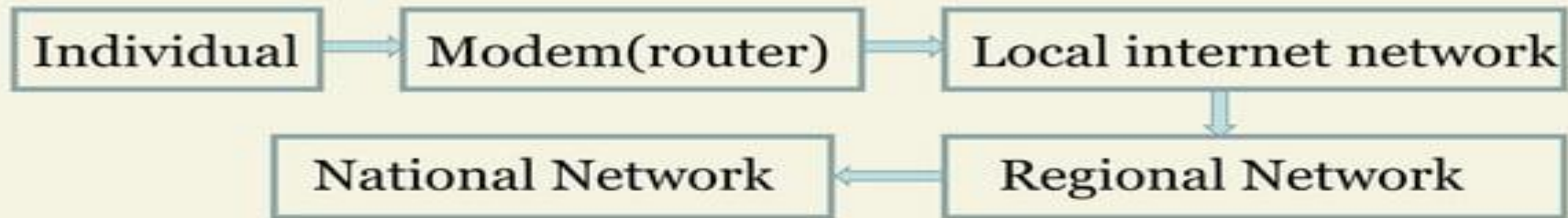
It is by definition a meta-network, a constantly changing collection of thousands of individual networks intercommunicating with a common protocol.

A short form of the compound word "inter-networking". This architecture is based in the very specification of the standard TCP/IP protocol, designed to connect any two networks which may be very different in internal hardware, software, and technical design.



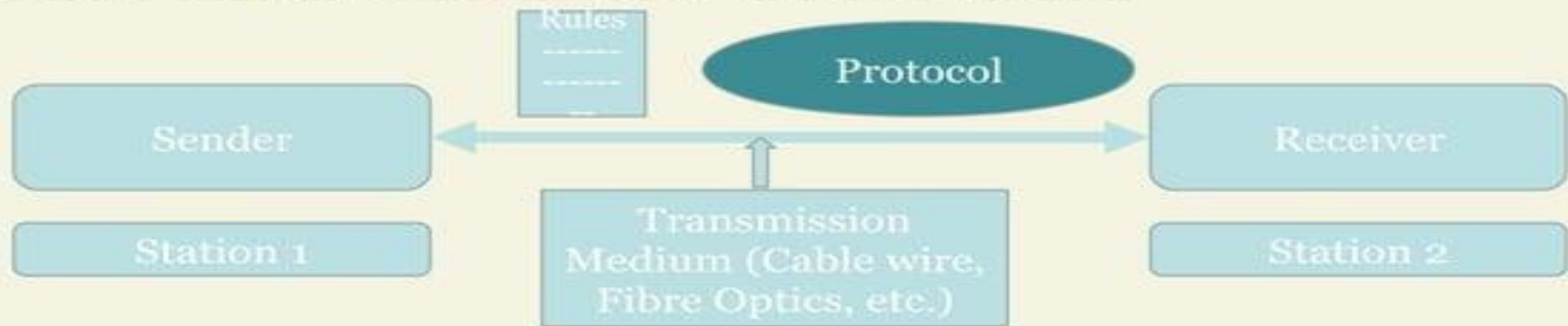
Internet Path

An individual's access to the Internet is often from home over a modem to a local Internet service provider who connects to a regional network connected to a national network.



Data Communication

- Data Communication means the exchange of between two or more devices via some transmission medium.
- Main components of data communication



Network Topology



Fully Connected Network Topology



Common Bus Topology



Mesh Network Topology



Star Network Topology

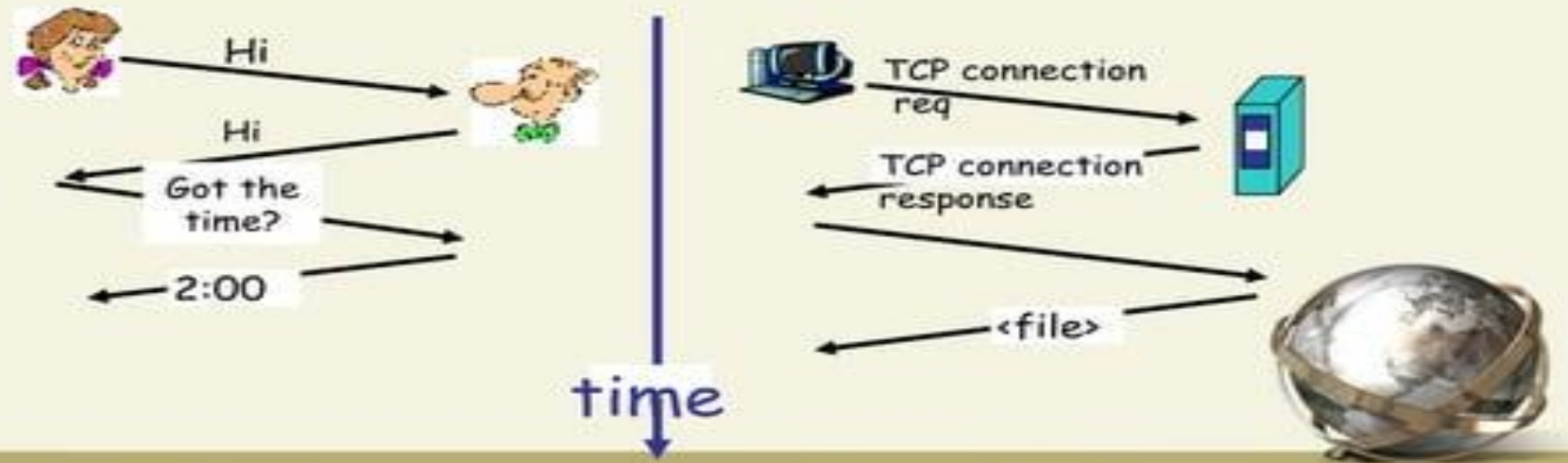


Ring Network Topology



Protocols

- End systems, routers, and other pieces of the Internet, run protocols that control the sending and receiving of information within the Internet.
- TCP (Transmission Control Protocol) and IP (Internet protocol) are two of the most important protocols in the Internet.
- The Internet's principal protocols are collectively known as TCP/IP Protocol Suite.



OSI layer Model

7 Layers

7. Application Layer - Cables
6. Presentation Layer - Ethernet
5. Session Layer - IP
4. Transport Layer – TCP/UDP
3. Network Layer
2. Data Link Layer
1. Physical Layer

All
People
Seem
To
Need
Data
Processing



TCIP/IP Model

4 Layers

4. Application Layer

FTTP, HTTP,....

3. Transport Layer

TCP, VDP, SCTP

2. Internet Layer

ARP, RARP, ICMP, IGMP

1. Network Interface layer



Internet Layer

- Packaging
- Addressing
- Routing



IP Address

- 4 bytes
 - e.g. 163.1.125.98
 - Each device normally gets one (or more)
 - In theory there are about 4 billion available

An IP address serves two principal functions: host or network interface identification and location addressing.



Routing

- How does a device know where to send a packet?
 - All devices need to know what IP addresses are on directly attached networks
 - If the destination is on a local network, send it directly there
- If the destination address isn't local
 - Most non-router devices just send everything to a single local router

A router receives chunk of information from one of its incoming communication link and forwards it to one of its outgoing communication link.



Continue...

- *Routers* (nodes) are devices on multiple networks that pass traffic between them
- Individual networks pass traffic from one router or endpoint to another
- Responsible for end to end transmission
- Sends data in individual packets
- Maximum size of packet is determined by the networks



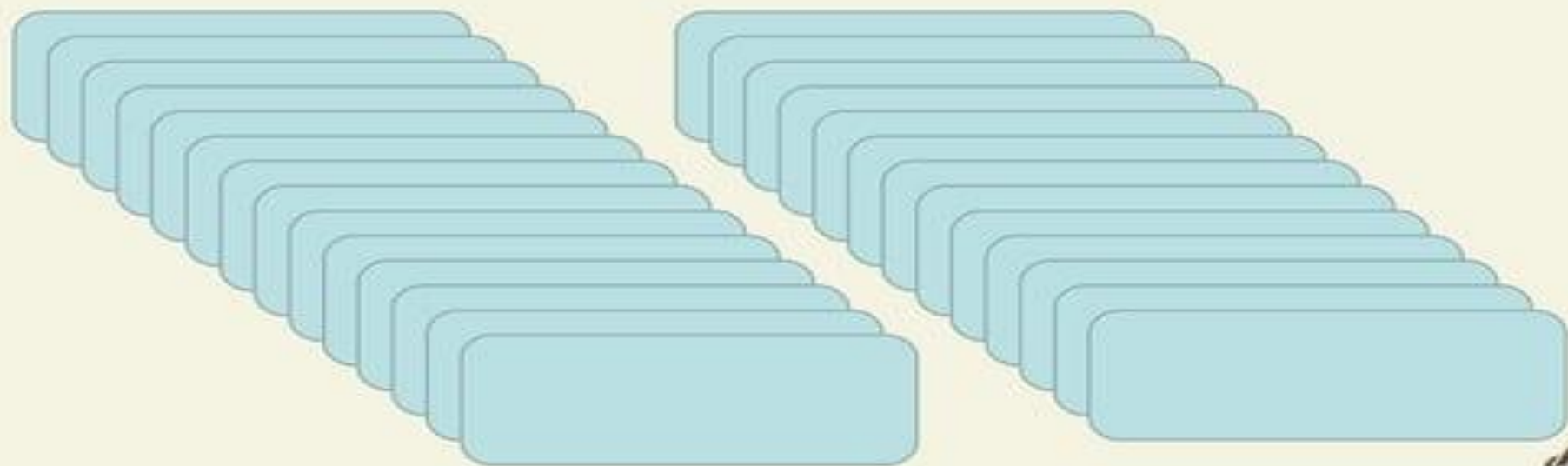
Terms to familiarize

- Domain name
- DNS Lookup
- Reverse DNS Lookup
- DNS Server
- HTTP

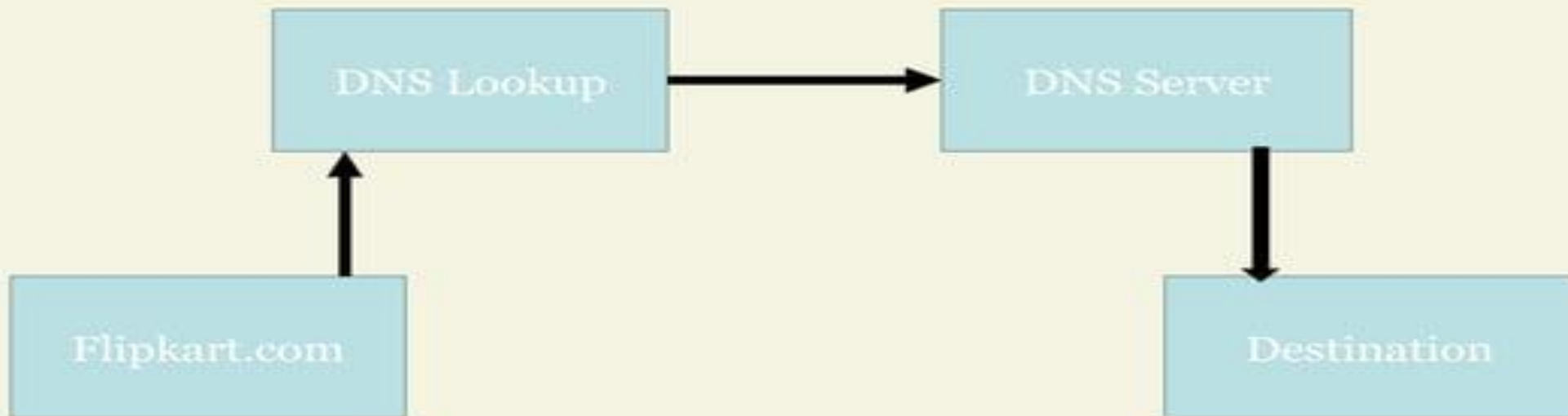


Interaction with Internet

- Suppose you type
<http://www.flipkart.com>

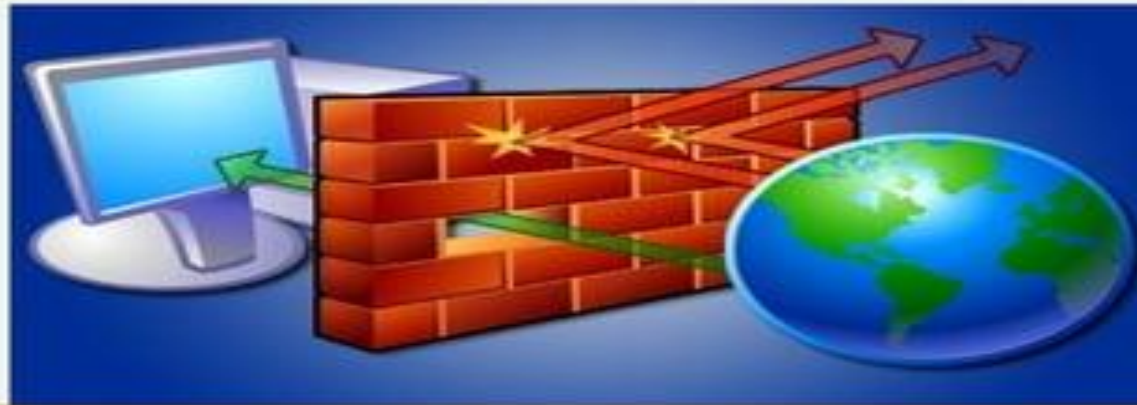


Working



Firewall

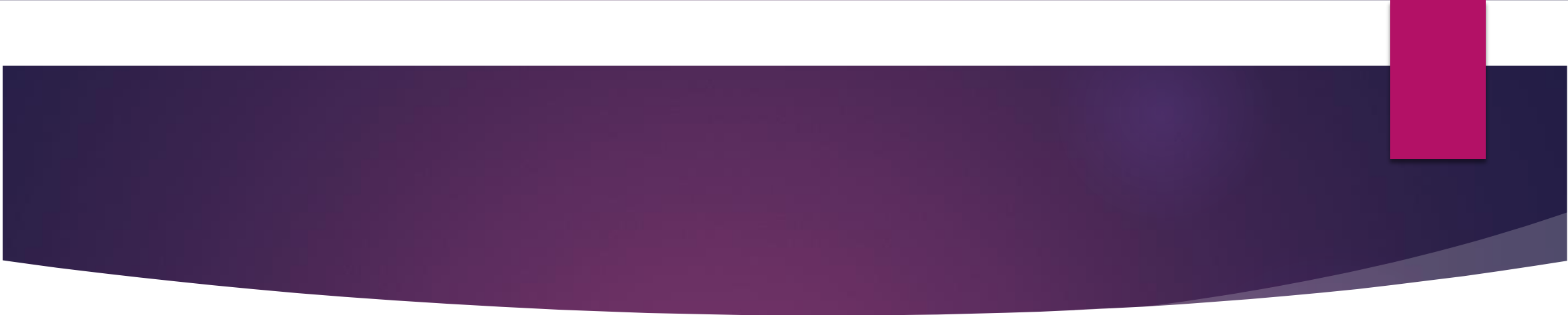
- A **choke point** of control and monitoring
- Interconnects networks with differing trust
- Imposes restrictions on network services
 - only authorized traffic is allowed
- Auditing and controlling access
 - can implement alarms for abnormal behavior
- Itself immune to penetration
- Provides **perimeter defence**



Food for thought!

- ✓ Internet vs. internet
- ✓ Difference between:
internet, intranet and extranet
- ✓ LAN vs. WAN
- ✓ Why Circuit Switching is not appropriate for
data traffic?





The End